

Chapter- 4

Marginal productivity theory of labor demand:-

The optimal level of employment for a firm depends on a ~~many~~ number of factors, such as, the cost of labor, the productivity of the work force, the level of production, and the price of the product. The theory of labor demand organizes these diverse considerations into a model.

The model:-

The standard model of labor demand is the marginal productivity theory of demand.

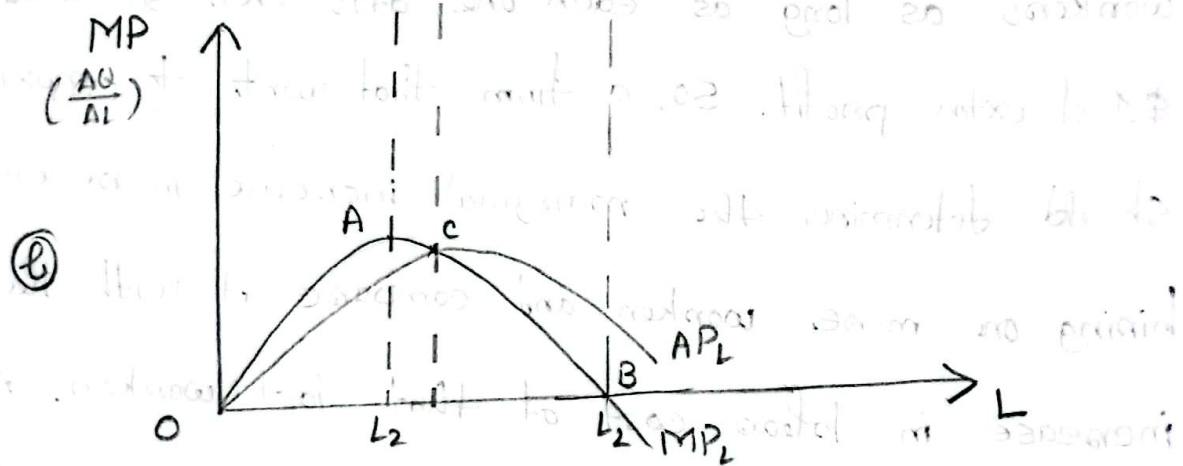
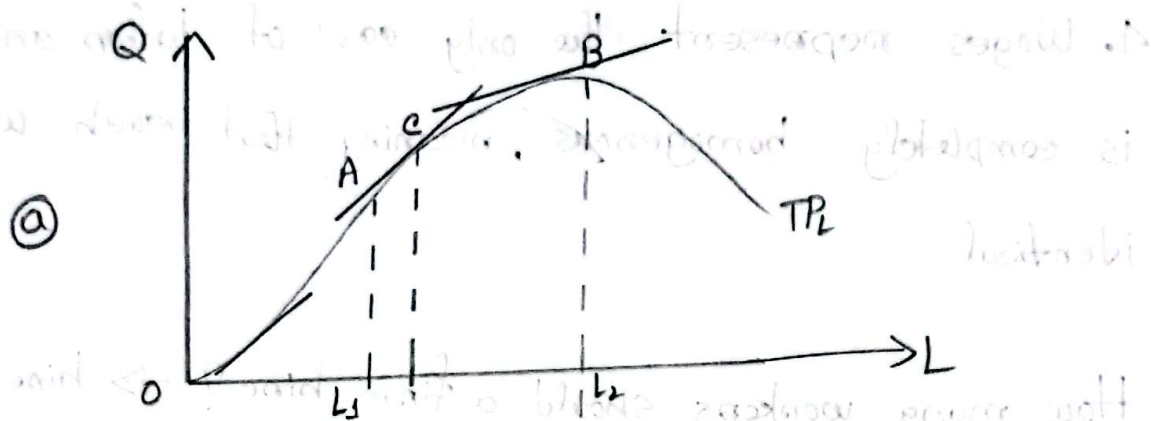
Assumptions:-

1. The goal of firms is to maximize profit.
2. Firms use only two factors of production, K and L.
3. Perfectly competitive product and labor market.

4. Wages represent the only cost of labor and labor is completely homogenous, meaning that, each worker is identical.

How many workers should a firm hire? $\rightarrow$ hire additional workers as long as each one adds even so much as \$1 of extra profit. So, a firm that wants to maximize profits should determine the marginal increase in revenue from hiring one more worker and compare it with the marginal increase in labor cost of that last worker. As long as the additional worker brings more revenue than cost, the firm adds to its profit by increasing employment.

To find out the equilibrium level of employment, we have to follow several steps.



⇒ The production function:-

$$Q = f(K, L)$$

This function indicates that the amount of output Q is a function of the amount of K and L produced by the firm, given the current state of technology. In the short run, K is fixed, so to increase output the firm has to hire additional workers. So, using

the production function equation, by holding K constant, we can predict the increase in output due to the increase in workers. This is also shown in figure (a), the total product curve TP_L , shows the amount of output at each level of employment.

⇒ The marginal and average product of labor:-

The marginal productivity of labor on MP_L shows the increment in production contributed by each additional worker hired. So,

$$MP_L = \frac{\Delta Q}{\Delta L}$$

The average product of labor measures the average amount of output produced per worker.

$$AP_L = \frac{Q}{L}$$

The graphs of MP_L and AP_L are shown in the figure

(b). Now, let's see the relationship between the total and marginal schedules.

The marginal product of each labor is given by the slope of a line drawn tangent to the total product curve TP_L . Up to L_1 the tangent lines become steeper, indicating that production is in the area of increasing returns. After L_1 , the tangent lines become flatter, indicating that diminishing returns have set in. Beyond L_2 the MP_L becomes negative. Plotting the slope of the tangent line in graph (b) yields the marginal product schedule MP_L . After point A, labor is subject to diminishing marginal productivity.

The maximum of the AP_L is at point C, where the slope of ray from the origin to the TP_L curve is the greatest.

$\Rightarrow$ Marginal Revenue Product:-

Marginal revenue is defined as the change in total revenue (TR) received by the firm from producing one more unit of output. That is,

$$MR = \frac{\Delta TR}{\Delta Q}$$

For a perfectly competitive market,

$$P = MR$$

Since a competitive firm can sell every additional unit of production at the same price, the increment of revenue from selling one more unit (MR) is exactly equal to the price it is sold for.

The marginal increase in revenue for this firm from hiring additional workers is known as the marginal revenue product (MRP_L) of labor. Calculated by,

$$MRP_L = MR \cdot MP_L$$

⇒ The marginal cost of labor:-

MRP_L shows the MB of a firm from hiring additional workers. The decision rule is to hire labor as long as the MB exceeds MC from hiring labor. The cost per hour to the firm from hiring each worker is equal to the wage rate (w).

$\Rightarrow$ The equilibrium level of employment:-

The equilibrium condition for a firm's labor demand is that the worker contributes just as much to revenue as he or she costs. and so, for a competitive firm,

$$w = MRP_L$$

$$\Rightarrow w = MR \cdot MP_L$$

$$[\because MRP_L = MR \cdot MP_L]$$

$$\Rightarrow w = P \cdot MP_L$$

$$[\because MR = P]$$

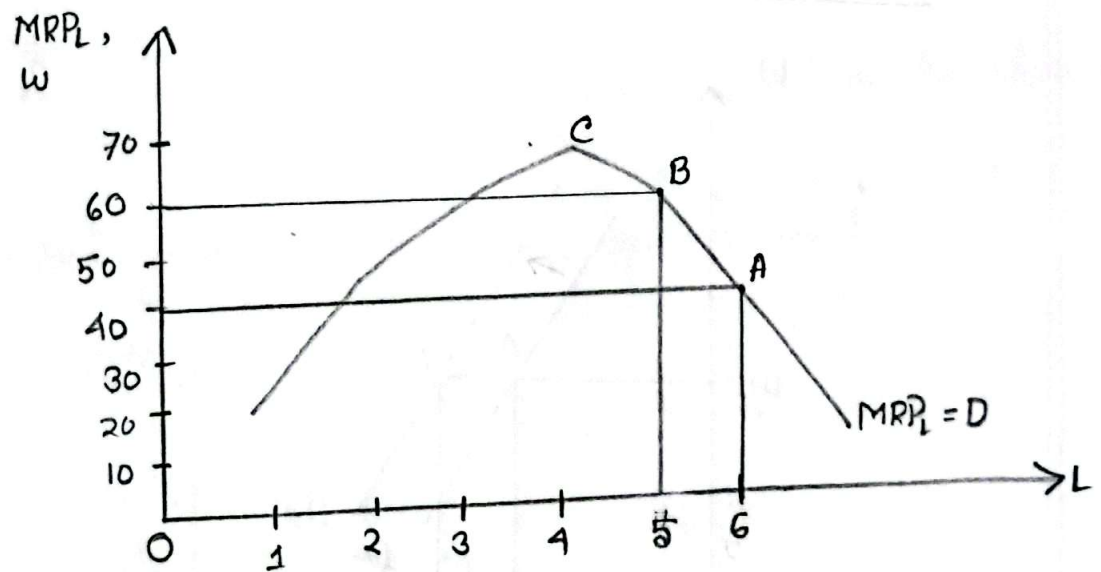
$$\Rightarrow \frac{w}{P} = MP_L$$

This equilibrium equation states that, a firm should continue to hire labor as long as the MRP_L of each successive worker exceeds the money wage rate. In other words,

The firm will hire workers as long as the MP_L is greater or equal to the real wage $(\frac{w}{P})$.

Short-run demand curve for labor:-

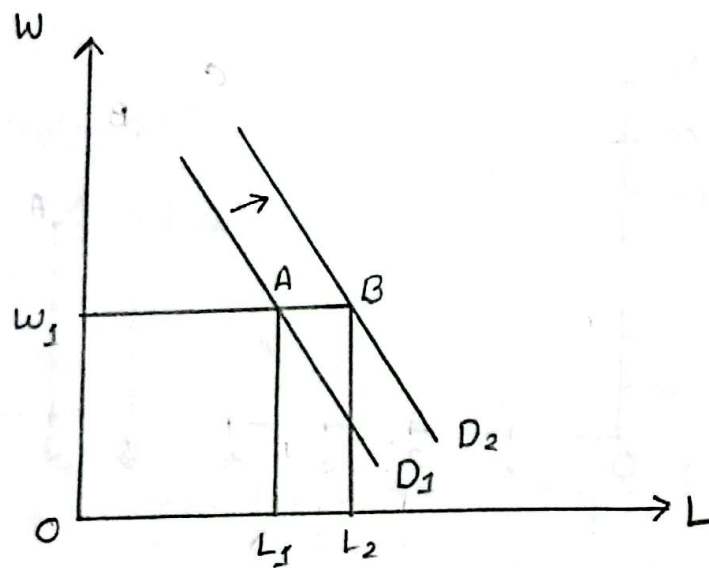
The short-run labor demand curve depicts the relationship between wage rate and the firm's desired level of employment, holding capital and all other factors constant.



The MRP_L schedule shows the additional revenue the firm obtains from hiring each worker. In the graph, the 5th worker brings in \$60, the 6th worker \$40 and so on. The downward sloping portion of the MRP_L is also the firm's short-run demand curve for labor, starting from point C. Thus, if the wage $w = \$40$, the firm would maximize profit by hiring 6 workers (Point A), since each of these

workers has an MRP_L at least as great as the wage. At a wage of \$60, the optimal employment is 5 workers (at point - B).

$\Rightarrow$ Change in product demand:-

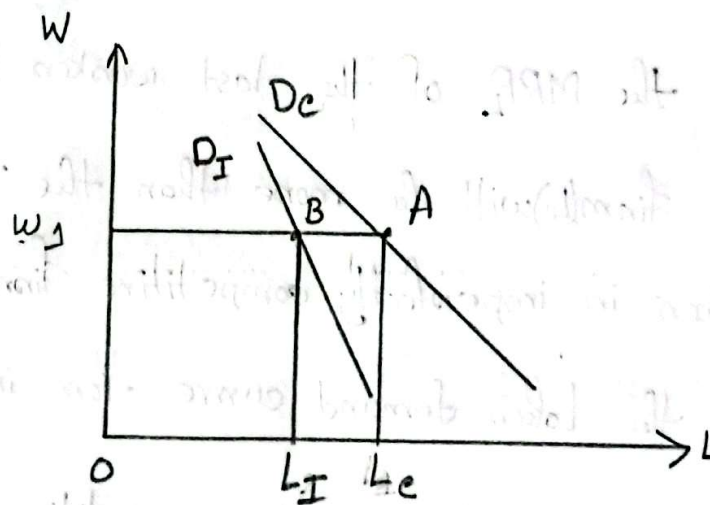


An increase in the product demand will cause the price of the product to rise in the market, ~~so the product~~

This immediately affects the demand for labor in each firm of the industry, since price P is part of the MRP_L calculation. A higher price, the MRP_L of each worker increases, so the labor demand curve shifts

rightward from D_1 to D_2 . At the wage of W_1 , employment can be expanded from L_1 to L_2 (Point A to B). ^{At} the higher product price, MRP_L of additional workers hired beyond L_1 will now exceed the wage cost W_1 , until employment reaches to L_2 . A decrease in product demand will do the opposite.

$\Rightarrow$ Imperfect competition in the product market:-



A competitive firm has a horizontal product demand curve and can sell all its output at a constant price.

An imperfectly competitive firm, faces a downward sloping product demand curve, implying that to sell more output

it has to lower the price. In this situation, the MR from selling a unit of output is less than the price it is sold for. If the imperfectly competitive firm wants to sell additional units of output, it has to further lower the price, causing the MR to decline further.

In imperfectly competitive market,

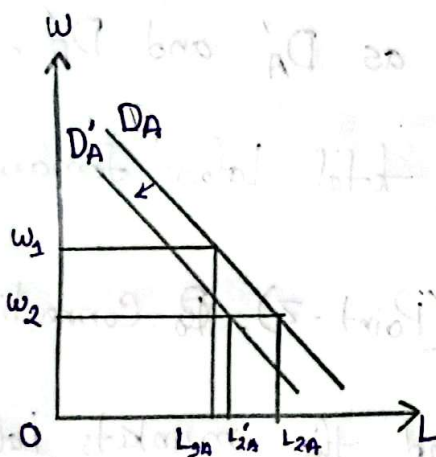
$$P > MR$$

So, firstly, the MRP_L of the last worker hired in the competitive firm (L_C) will be more than the MRP_L of the same worker in imperfectly competitive firm (L_I). This will cause the labor demand curve for imperfectly competitive firm (D_I) to lie behind the competitive one (D_C). So, at wage w_1 , the employment in imperfectly competitive firm (L_I) will be less than competitive one (L_C).

Secondly, in imperfectly competitive firm, not only the demand for labor is reduced but also demand curve

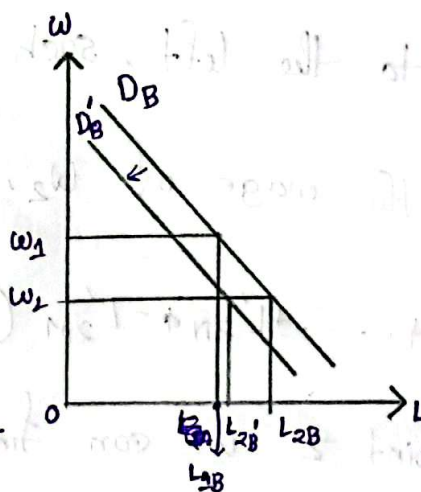
D_I is steeper than D_C . The reason behind this is, the MRP_L^I declines for two reasons - MRP_L^I of ~~the~~ each worker declines due to diminishing returns in production and MR^I from extra production also declines.

Market labor demand curve:-



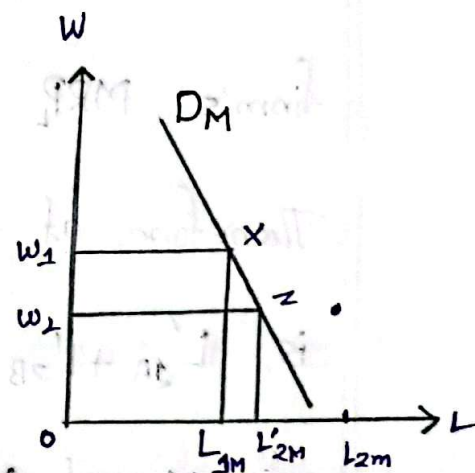
Firm - A

(a)



Firm - B

(b)



The market

(c)

The lines D_A and D_B in graphs (a) and (b) show the labor demand curve for each individual firm. At the wage rate of w_1 , total employment in the market (graph-c) is $L_{1A} + L_{1B} + \dots + L_{1N} = L_{1M}$ (Point-X). A fall in the

wage from w_1 to w_2 in any one of the firm would cause a downward shift of the ~~MRP~~ MRP_L schedule, increasing employment from L_{1A} to L_{2A} in firm A, for example. If the wage falls to w_2 for all firms, the resulting expansion in employment and output will cause a decline in the product's price, shifting each of the firm's MRP_L to the left, such as D_A' and D_B' .

Therefore, at the wage of w_2 , total labor demand is, $L'_{1A} + L'_{2B} + \dots + L'_{2N} = L'_{2M}$ (Point-Z). ~~By~~ Connecting point X and point Z we can find the market's labor

demand curve D_M .

Criticism of the theory of labor demand

The MP theory of labor demand can predict both the optimal level of employment for a firm and how this employment level will change in reaction to various economic events. But, this theory is subjected to various criticisms, 5 of these criticisms are:-

① Limits to human cognition:- One objection is that the information and computations necessary to operationalize the theory generally exceed the mental and cognitive ability of most employers and managers. For example, while the concept of MP is clear in theory, can a typical manager of a firm can measure or even approximate the likely increase in production from hiring a new employee? Even if a MP can be calculated, how can the firm meaningfully compare a worker's MRP_L with the W , when product prices (P) and

output levels (Q) are constantly changing.

Counters:-

1st, Although the managers may not consciously use marginal calculations, the decisions they reach must approximate those predicted by the theory for the firm to survive in the competitive world.

2nd, though the managers will not be able to calculate the marginal contribution of each individual worker on adjust employment, they can identify revenues and costs associated with the particular activity and then adjust the employment in light of their contribution to profit.

② Nonmaximizing behavior:- The 2nd criticism is about the assumption of profit maximization. For a firm, the calculations of MRP_L vs MC of labor is the goal of maximizing profit.

In the business firms, where the ownership and control are separated, are more accurately characterized as satisfying with respect to profit. In this view, the managers strive

to achieve the minimum level of profit in order to protect their survival at the firm. Once the minimum goal is achieved, the managers ~~do~~ tolerate some slack in the operation of the firm and no longer strive to fully minimize production costs.

Counteris:-

The ultimate goal of each firm is survival and the survival in a competitive economy requires keeping costs and employment down and profits up, because of that the single goal of maximizing profit is subordinated. For a firm's ϕ survival and growth requires conscious efforts to minimize cost.

③ Fixed capital/Labour ratio:- The nature of technology makes it impossible to derive a continuous MP_L schedule. The MP_L schedule assumes that the fixed K can be spread among greater and greater numbers of workers as employment increases. However, many types of production processes requires fixed K/L ratio. Because of this situation, it is impossible

either to attribute a unique MP_L to each individual worker or to derive a continuous MP_L schedule.

Counter:-

The proponents of MP theory discount the seriousness of the objections. They think that, in nearly all real-world production situations there is no fixed, necessary relationship between K and L . They also argue that more often the fixed K/L is stemmed from union work rules or regulatory constraints rather than from technology itself.

④ Increasing return to labor:- Labor may be subject to increasing returns in the short run not diminishing returns as the theory assumes. The diminishing return theory assumes, as the employment increases, the MP_L should decline as the fixed K is spread over more workers, so the MRP_L and D_L schedule are downward sloping, for a competitive firm. Numerous empirical studies have found that, the Labor

productivity varies directly with the level of employment in the firm - when employment rises, business cycle goes upward, output rises. So, the $AP_L (Q/L)$ rises with increased employment, suggesting that, $MP_L (\Delta Q/\Delta L)$ also increases, implying that MRP_L schedule is actually upward sloping.

Counters:-

1st, statistical illustration caused by faulty data on the level of production.

2nd, The MP theory assumes that workers put in a constant level of effort, while in reality the effort is likely to slacken in bad times and increases in good times.

⑤ Independence between wage and workers productivity:-

The final criticism is about the relationship between wage rate and the productivity of the workers. The theory assumes that, The MP schedule is solely determined by the technology of

production and the size of the firm's capital stock. A change in the wage will cause the firm to move up or down the MRP_L curve, but the curve itself will not change. The critics argue that, in real life the worker's productivity is likely to be directly related to the wage. One reason is that, higher wage allows workers to improve their physical ability to work through improved nutrition and health; a second reason can be higher wage ~~increase~~ is likely to stimulate greater worker effort and higher morale among employees. So, they say that, there is a separate MRP_L for each level of wage. Because of this, employment will increase or decrease in the response of wage change can no longer be unambiguously predicted.

Counters:-

1st, They admit that, the worker's productivity is dependent on the wage rate, at to some degree in underdeveloped countries. In developed economies, modest adjustment in wage

rates will probably have small negligible impacts on workers' performance.

2nd, considerable disagreement among specialists in organizational behavior over the extent to which pay really influences performance at the firm level, can be positive or negative or no relationship at all.

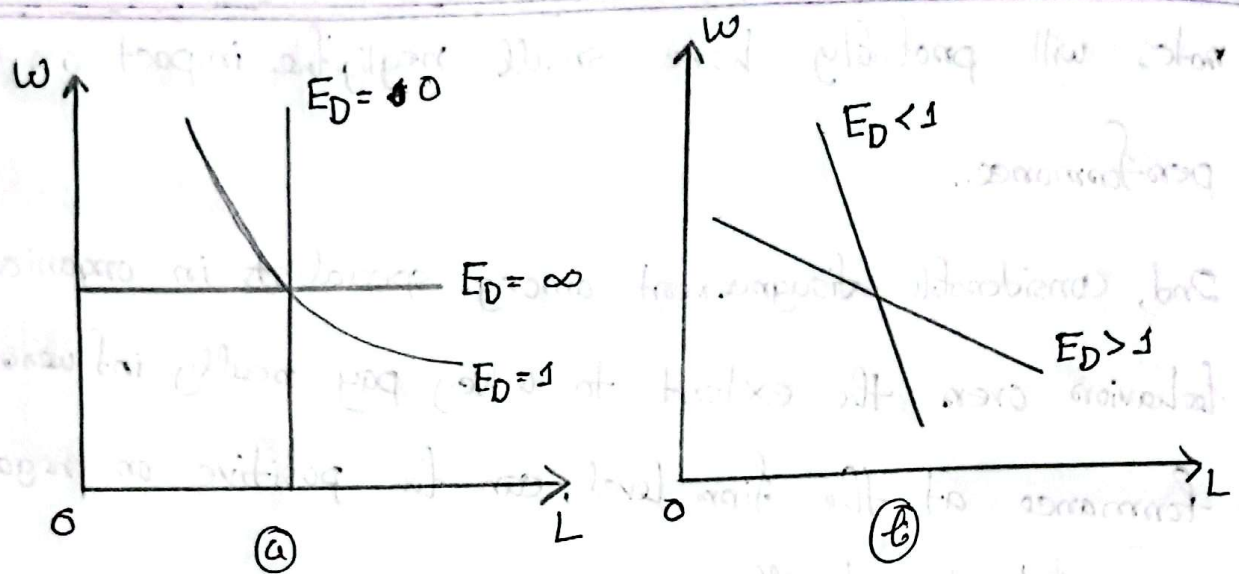
The elasticity of demand for labor

The downward sloping labor demand curve shows that wage rate and level of employment are inversely related. In elasticity of labor demand, we discuss how sensitive employment is to change in the cost of labor:-

$$E_D = \frac{\% \Delta L}{\% \Delta W}$$

(-)

as L and W inversely related.



① $E_D = 0$, as an increase in w , say 10% ($\Delta \% w = 10\%$), causes no change in the demand for labor ($\Delta \% L = 0$).

So, the demand curve is perfectly inelastic, which is shown in figure (a).

② $E_D < 1$, when E_D is less than 0 but greater than 1, the demand will be inelastic. In this case, suppose an increase in w of 10% ($\Delta \% w = 10$) will cause only 5% increase in L_D ($\Delta \% L = 5$), here $E_D = 0.5$. This shows that, the labor demand is relatively insensitive to labor cost. The L_D curve is shown in figure (b).

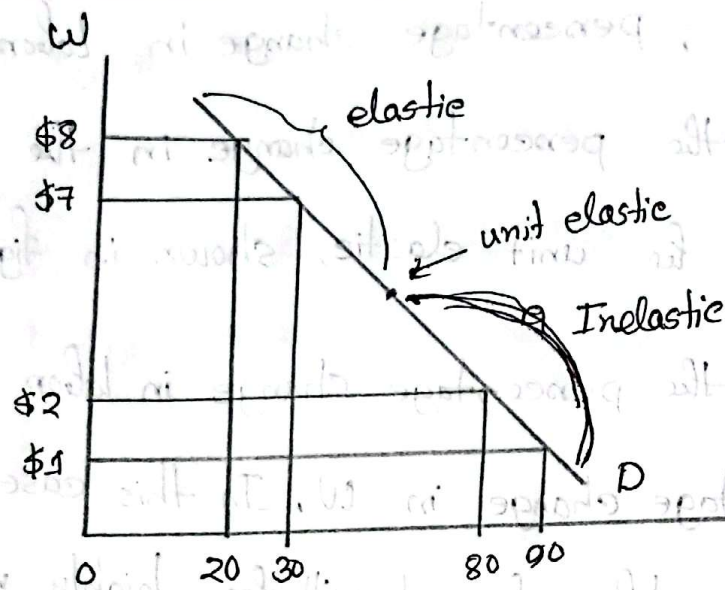
③, $E_D = 1$, percentage change in labor demand is just equal to the percentage change in the w . Demand in this case will be unit elastic, shown in figure (a).

④ $E_D > 1$, the percentage change in labor demand exceeds the percentage change in w . In this case, the demand is elastic, the labor demand will be highly responsive to change in w . The demand curve is shown in figure (b).

⑤ $E_D = \infty$, the firm is willing to hire all the additional employees it can at the prevailing wage, but ^{will} hire no employees at any higher wage, demand is known as perfectly elastic, shown in figure (a).

Special relationships:-

① The relationship between elasticity and the slope of a linear demand curve. Flat demand curves are said to be elastic, strictly speaking, this is incorrect. Any downward sloping straight line will have both elastic and inelastic portion.



In the graph, if the wage falls from \$8 to \$7, the employment will rise from 20 to 30, hence $E_D = 4$ ($50\% / 12.5\%$), demand is very elastic. If the wage falls from \$2 to \$1, the employment will rise from 80 to 90, $E_D = 0.125$ ($12.5\% / 50\%$), which shows demand is quite inelastic. So, for the ^{same} \$1 change in the w , demand on the top half of the curve is elastic, the mid point is unit elastic and the bottom half is inelastic.

② A firm's wage bill is the total money outlay for labor.

Hence, Wage bill = $W \cdot L$

If the labor demand is inelastic, a rise in w will result in a net increase in the total wage bill, as the ^{increase in} $\% \Delta W$ will exceed the decline in $\% \Delta L$, causing the $W \cdot L$ to be greater than before. The opposite is true for a decrease.

But if the L_D is elastic, a rise in w will result in a net decrease in the total wage bill as, $\% \text{increase in } w < \text{decrease in } L \text{ employment}$.

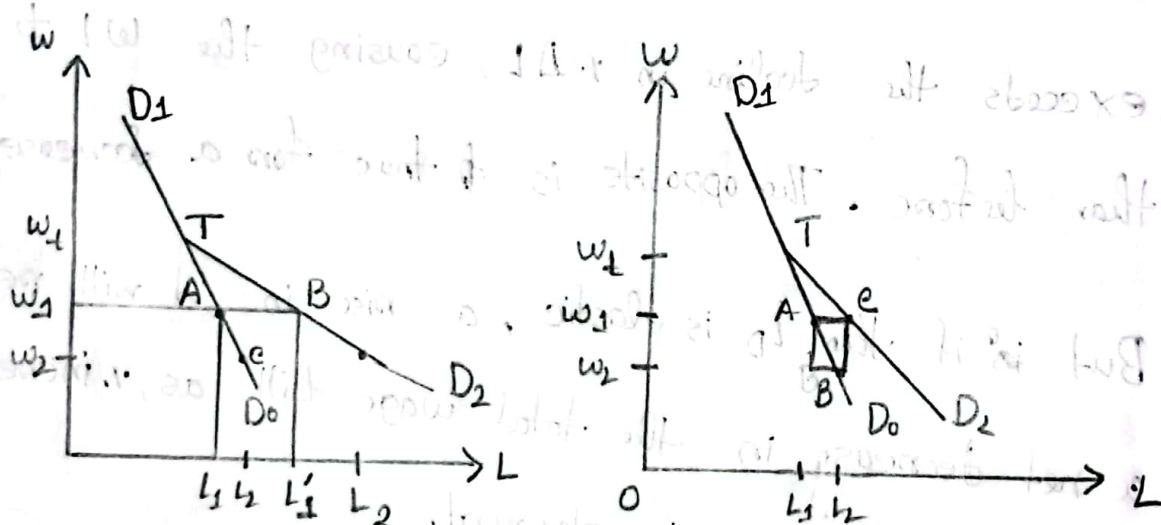
③ Why L_D is elastic in some cases and inelastic in others.

A rise in w in the market raises the cost of production, leading to a reduction in employment for two reasons.

First, the firms will be forced to raise their production price, resulting fewer sales, less production and less L_D . Second, higher cost of labor will motivate firms to substitute capital

in the ~~lab~~ long run, leading to a further reduction in the D_L , so more elastic the labor demand will be.

Employment effect of wage subsidy program



A wage subsidy program aims to create additional jobs in the economy by subsidizing the employer's cost of labor. It focuses on the private sector.

In designing wage subsidy program, policymakers are faced with some conflicts. They must weigh the trade-off between program costs and the size of the financial incentive offered to firms to increase employment. For

solving this dilemma, there are different types of wage subsidy programs:-

- 1) Categorical program:- only applies to identifiable low-skill group, such as teenagers and welfare recipients, who are hard-core unemployed.
- 2) Only for people newly hired, not for existing employees, known as marginal or incremental wage subsidy.
- 3) New workers who earn a wage less than the target wage (w_t) qualify for a subsidy;
~~so, $w_t < w_p$~~
- 4) For every worker hired, the employer receives a subsidy payment per hour, S , equal to some percent, n , of the difference between the target wage (w_t) and hiring wage (w_p).

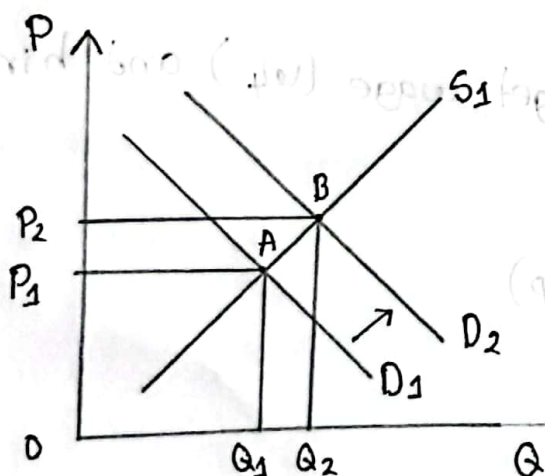
$$\text{so, } S = n (w_t - w_p)$$

In the graph, the D_0D_1 is the before subsidy demand curve. Here, wage is w_1 , a typical firm would hire L_1 workers (point-A); at a wage of w_2 , employment will be L_2 (point - c).

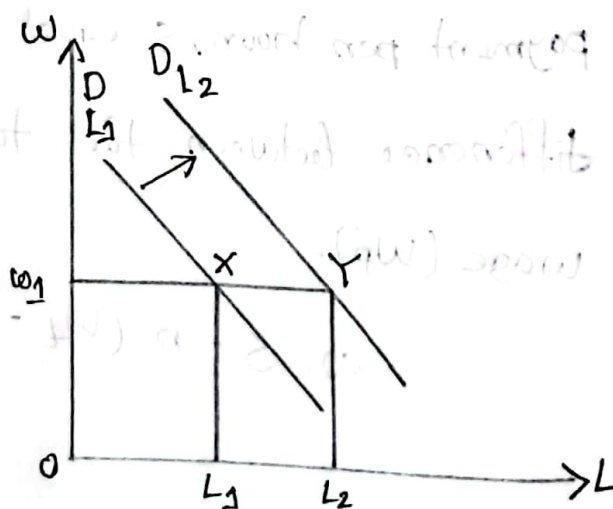
After the wage subsidy program demand curve will be D_1TD_2 (in graph @). After the program, for w_1 wage the employment will be L_1' , shown as point B at the D_1TD_2 .

At a lower wage of w_2 , a larger increase in employment from L_2 to L_2' (Point c to E) due to the larger subsidy payment per workers hired.

Import restriction and employment



Product market



Labor market

Increasing import restriction reduces competition in any market driving up the product price. When the product price is higher, MRP_L of the industry is higher, leading to a higher level of employment. Whatever the form of import restriction (tariff, quota etc.) is, the impact on the auto industry labor market will be the same.

In the graph, without import restrictions, the demand curve for domestically produced cars is assumed to be D_1 and the supply curve is S_1 . The equilibrium price of domestic cars is P_1 , production is Q_1 (Point A). Graph (b), shows the demand curve for labor is D_{L_1} , employment of L_1 (Point X).

Imposition of import restriction, ~~the~~ demand of domestically produced cars will increase, shifting the demand curve for domestic cars from D_1 upward to D_2 . At the new equilibrium point B, output increases from Q_1 to Q_2 , price increases from P_1 to P_2 . Given the increase in sales from Q_1 to Q_2 ,

the industry's demand for labor would also increase. In graph (b), the increase in sales will shift the labor demand curve from D_{L1} to D_{L2} . At the prevailing union wage of w_1 , employment will increase from L_1 to L_2 (point X to Y).

This simple model predicts that import restrictions would lead to higher output, higher prices and higher employment in the domestic car industry.